

Q1: Hydrogen is the first element in the periodic table and is the smallest among all elements present on the table. Its atomic number is 1 which indicates that it has only one electron that is orbiting in its valance shell. It has one shell only and is the lightest element present on the periodic table. In the modern periodic table, the position of the elements is largely dependent on their electronic configuration and the electronic configuration of hydrogen is 1. If hydrogen gets rid of its electron then it can attain the electronic configuration of the noble gasses. This character of the hydrogen element is similar to that of the alkali metals. But if hydrogen gains the one electron then its configuration will be similar to that of halogens. However, the properties of hydrogen do not completely match any of these groups. Due to this reason, the position of hydrogen is still considered as undecided as hydrogen behaves like halogens as well as like alkali metals.

Best position of hydrogen in the periodic table is above I.A Group which is mainly due to

- (A) Both are electropositive
- (B) Similar outer most shell electronic configuration
- (C) Both form ionic compounds
- (D) All

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- (A) Both are electropositive
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Solution:

Hydrogen is placed in group I of the modern periodic table due to following reasons:-

- (1) It shows outer electronic configuration similar to group I elements (i.e. $1s^1$).
- (2) It readily loses its electron to form H^+ ion which resembles the electropositive character of group I elements.
- (3) It shows +1 oxidation state like alkali i.e. group I elements.
- (4) Hydrogen also acts as a strong reducing agent like alkali metals and also forms binary compounds with electronegative elements which resembles the group I elements.

Q2: Hydrogen is the first element in the periodic table and is the smallest among all elements present on the table. Its atomic number is 1 which indicates that it has only one electron that is orbiting in its valence shell. It has one shell only and is the lightest element present on the periodic table. In the modern periodic table, the position of the elements is largely dependent on their electronic configuration and the electronic configuration of hydrogen is 1. If hydrogen gets rid of its electron then it can attain the electronic configuration of the noble gases. This character of the hydrogen element is similar to that of the alkali metals. But if hydrogen gains the one electron then its configuration will be similar to that of halogens. However, the properties of hydrogen do not completely match any of these groups. Due to this reason, the position of hydrogen is still considered as undecided as hydrogen behaves like halogens as well as like alkali metals.

The usual valence of hydrogen is:

- (A) -2
- (B) 0
- (C) +2
- (D) -1 or some times +1

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The usual valence of hydrogen is:

- (A) -2
- (B) 0
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- (D) -1 or some times +1

Solution:

Hydrogen can either gain or lose an electron to have a stable electron shell, giving it a valence of +1 or -1.

Q3: Hydrogen is the first element in the periodic table and is the smallest among all elements present on the table. Its atomic number is 1 which indicates that it has only one electron that is orbiting in its valance shell. It has one shell only and is the lightest element present on the periodic table. In the modern periodic table, the position of the elements is largely dependent on their electronic configuration and the electronic configuration of hydrogen is 1. If hydrogen gets rid of its electron then it can attain the electronic configuration of the noble gasses. This character of the hydrogen element is similar to that of the alkali metals. But if hydrogen gains the one electron then its configuration will be similar to that of halogens. However, the properties of hydrogen do not completely match any of these groups. Due to this reason, the position of hydrogen is still considered as undecided as hydrogen behaves like halogens as well as like alkali metals.

The element symbol for hydrogen is :

- (A) K
- (B) H
- (C) He
- (D) Hn

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The element symbol for hydrogen is H.

Q4: Hydrogen is the first element in the periodic table and is the smallest among all elements present on the table. Its atomic number is 1 which indicates that it has only one electron that is orbiting in its valance shell. It has one shell only and is the lightest element present on the periodic table. In the modern periodic table, the position of the elements is largely dependent on their electronic configuration and the electronic configuration of hydrogen is 1. If hydrogen gets rid of its electron then it can attain the electronic configuration of the noble gasses. This character of the hydrogen element is similar to that of the alkali metals. But if hydrogen gains the one electron then its configuration will be similar to that of halogens. However, the properties of hydrogen do not completely match any of these groups. Due to this reason, the position of hydrogen is still considered as undecided as hydrogen behaves like halogens as well as like alkali metals.

What is the electronic configuration of hydrogen?

- (A) $1s^1$
- (B) $1s^2$
- (C) $2s^1$
- (D) $2s^2$

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- (A) $1s^1$
- (B) $1s^2$
- (C) $2s^1$
- (D) $2s^2$

Solution:

The electronic configuration of hydrogen is $1s^1$, as it has only one electron. It belongs to the first group and is ready to gain one electron in order to become a noble gas like helium which is next to it.

Q5: A hydrogen atom is extremely reactive in its original form. It has one valence electron in its only shell, which makes it highly reactive and unstable. So hydrogen combines with another atom of hydrogen to give us Dihydrogen. We denote this as H_2 . This molecular form of hydrogen is the most common form of hydrogen on earth. Since the molecule is neutral (has no charge) and its orbit is now complete it is a stable gas. Actually, a dihydrogen molecule (H_2) is the smallest molecule on the planet. Hydrogen has three naturally occurring isotopes, namely Protium (1H), Deuterium (2H) and Tritium (3H). Of these, tritium is the only one not stable, it is in fact radioactive. The difference between the three is the number of neutrons.

Heavy hydrogen is used for which of the following?

- (A) Filling up balloons
- (B) Studying reaction mechanism
- (C) Oxidizing agent
- (D) All of the above.

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- (A) Filling up balloons
- (B) Studying reaction mechanism
- (C) Oxidizing agent
- (D) All of the above.

Solution:

We use heavy hydrogen to study reaction mechanism as it is very easy to identify deuterium.

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The nuclei of tritium (H^3) atom would contain neutrons equal to:

- (A) 1
- (B) 2
- (C) 3
- (D) 4

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Solution:

The atomic number and mass number of Tritium are 1 and 3 respectively. So it contains 1 electron, 1 proton, and 2 neutrons.

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Reactivity of hydrogen is _____ compared to halogens.

- (A) high
- (B) low
- (C) zero
- (D) negative

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Reactivity of hydrogen is _____ compared to halogens.

- (A) high
- (B) low
- (C) zero
- (D) negative

Solution:

Though hydrogen has many properties of halogens, in terms of reactivity it shows very low similarity when compared to halogens. This is one of the reasons, it's probably not placed in halogens, but in alkali metals.

Q8: A hydrogen atom is extremely reactive in its original form. It has one valence electron in its only shell, which makes it highly reactive and unstable. So hydrogen combines with another atom of hydrogen to give us Dihydrogen. We denote this as H_2 . This molecular form of hydrogen is the most common form of hydrogen on earth. Since the molecule is neutral (has no charge) and its orbit is now complete it is a stable gas. Actually, a dihydrogen molecule (H_2) is the smallest molecule on the planet. Hydrogen has three naturally occurring isotopes, namely Protium (1H), Deuterium (2H) and Tritium (3H). Of these, tritium is the only one not stable, it is in fact radioactive. The difference between the three is the number of neutrons.

Which of the following is not an isotope of hydrogen?

- (A) helium
- (B) protium
- (C) deuterium
- (D) tritium

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Which of the following is not an isotope of hydrogen?

- (A) helium
- (B) protium
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Solution:

Hydrogen exists in the form of three isotopes namely protium, deuterium and tritium. The atomic numbers of all the three isotopes of hydrogen are same i.e. 1, but their symbols, relative atomic mass, density, etc vary.

Q9: Dihydrogen is used as a fuel as it releases immense heat on combustion. This energy released comes around three times more than petrol when it comes to mass for mass basis. It is also observed that pollutants released with the combustion of hydrogen are less than petrol thus the only ones left will be oxides of dinitrogen and they can be decreased with the addition of a little water into the cylinder. This will lessen the temperature which will lead to the absence of reaction between dinitrogen and dioxygen. But, we need to consider the mass of dihydrogen container in such a case. Dihydrogen gas can also be converted to liquid form through cooling it to 2K but this would need costly insulated tanks which have posed as a limitation for the researcher to resort for alternative techniques for utilising dihydrogen. Thus, the Hydrogen Economy can work as a great alternative in such a case as it transports and stores energy in a liquid or gaseous form of dihydrogen.

What is the problem that arises with hydrogen as a fuel?

- (A) storage
- (B) production
- (C) process
- (D) availability

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What is the problem that arises with hydrogen as a fuel?

- (A) storage
- (B) production
- (C) process
- (D) availability

Solution:

The production of hydrogen is very easy and its process is also simple because the availability is enormous in the atmosphere as well as in the form of water, but the problem comes in the storage because we need to store hydrogen in a high-pressure tank or in a cryogenic tank.

Q10: Dihydrogen is used as a fuel as it releases immense heat on combustion. This energy released comes around three times more than petrol when it comes to mass for mass basis. It is also observed that pollutants released with the combustion of petrol are less than petrol thus the only ones left will be oxides of dinitrogen and they can be decreased with the addition of a little water into the cylinder. This will lessen the temperature which will lead to the absence of reaction between dinitrogen and dioxygen. But, we need to consider the mass of dihydrogen container in such a case. Dihydrogen gas can also be converted to liquid form through cooling it to 2K but this would need costly insulated tanks which have posed as a limitation for the researcher to resort for alternative techniques for utilising dihydrogen. Thus, the Hydrogen Economy can work as a great alternative in such a case as it transports and stores energy in a liquid or gaseous form of dihydrogen.

Is liquid hydrogen fuel dangerous?

- (A) yes
- (B) no
- (C) may be
- (D) may not be

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Is liquid hydrogen fuel dangerous?

- (A) yes
- (B) no
- (C) may be
- (D) may not be

Solution:

Yes, hydrogen fuel is dangerous as it has low ignition energy and high combustion energy. It tends to leak easily from tanks, this is a reason for many explosions at hydrogen fuel manufacturing or storing centers.

Q11: Dihydrogen is used as a fuel as it releases immense heat on combustion. This energy released comes around three times more than petrol when it comes to mass for mass basis. It is also observed that pollutants released with the combustion of petrol are less than petrol thus the only ones left will be oxides of dinitrogen and they can be decreased with the addition of a little water into the cylinder. This will lessen the temperature which will lead to the absence of reaction between dinitrogen and dioxygen. But, we need to consider the mass of dihydrogen container in such a case. Dihydrogen gas can also be converted to liquid form through cooling it to 2K but this would need costly insulated tanks which have posed as a limitation for the researcher to resort for alternative techniques for utilising dihydrogen. Thus, the Hydrogen Economy can work as a great alternative in such a case as it transports and stores energy in a liquid or gaseous form of dihydrogen.

Can dihydrogen be used as a Fuel?

- (A) yes
- (B) no
- (C) may be
- (D) cannot say

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- (A) yes
- (B) no
- (C) may be
- (D) cannot say

Solution:

Yes, dihydrogen can be used as a fuel and the process of usage of liquid hydrogen as an alternate source of energy is given by the term hydrogen economy. In the near future, there is a possibility of using liquid hydrogen as a fuel.

Q12: Dihydrogen is used as a fuel as it releases immense heat on combustion. This energy released comes around three times more than petrol when it comes to mass for mass basis. It is also observed that pollutants released with the combustion of petrol are less than petrol thus the only ones left will be oxides of dinitrogen and they can be decreased with the addition of a little water into the cylinder. This will lessen the temperature which will lead to the absence of reaction between dinitrogen and dioxygen. But, we need to consider the mass of dihydrogen container in such a case. Dihydrogen gas can also be converted to liquid form through cooling it to 2K but this would need costly insulated tanks which have posed as a limitation for the researcher to resort for alternative techniques for utilising dihydrogen. Thus, the Hydrogen Economy can work as a great alternative in such a case as it transports and stores energy in a liquid or gaseous form of dihydrogen.

What do you think the hydrogen economy involves?

- (A) mechanism
- (B) production
- (C) implementation
- (D) process

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What do you think the hydrogen economy involves?

- (A) mechanism
- (B) production
- (C) implementation
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Solution:

Hydrogen economy is a technology that involves production, transportation, and storage of energy in the form of liquid hydrogen as it is an alternate source of energy, there are many ways in which hydrogen can be used as fuel.

Q13: The bonding between hydrogen and other elements is always covalent. It just varies in its degrees of covalency. The general chemical formula of a hydride is MH_x . Here M is the other element with which hydrogen makes a binary compound. And the x denotes the number of hydrogen atoms. There are three possible types of hydrides based on their structure and chemical properties. Saline Hydrides form when hydrogen reacts with s-Block elements. These are elements found in Group 1 and Group 2. These alkali metals and alkaline earth metals are more electropositive than hydrogen. Covalent hydrides that form when hydrogen reacts with p-block elements. These are highly electronegative elements, far more than hydrogen. Hydrogen atoms form a covalent bond with these p-block atoms. Metallic Hydride are formed when hydrogen reacting with transition elements.

Assertion: The alkali metals can form ionic hydrides which contains the hydride ion.

Reason: The alkali metals have low electronegativity, their hydrides conduct electricity when fused and liberate hydrogen at the anode.

- (A) Assertion and Reason both are correct statements and reason is correct explanation for assertion.
- (B) Assertion and Reason both are correct statements but reason is not correct explanation for assertion.
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- (D) Assertion is wrong statement but reason is correct.

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Solution:

The alkali metals can form ionic hydrides which contains the hydride ion because the alkali metals have low electronegativity, their hydrides conduct electricity when fused and liberate hydrogen at the anode.

Q14: The bonding between hydrogen and other elements is always covalent. It just varies in its degrees of covalency. The general chemical formula of a hydride is MH_x . Here M is the other element with which hydrogen makes a binary compound. And the x denotes the number of hydrogen atoms. There are three possible types of hydrides based on their structure and chemical properties. Saline Hydrides form when hydrogen reacts with s-Block elements. These are elements found in Group 1 and Group 2. These alkali metals and alkaline earth metals are more electropositive than hydrogen. Covalent hydrides that form when hydrogen reacts with p-block elements. These are highly electronegative elements, far more than hydrogen. Hydrogen atoms form a covalent bond with these p-block atoms. Metallic Hydride are formed when hydrogen reacting with transition elements.

Assertion: Saline hydrides are non-volatile, non-conducting and crystalline solids.

Reason: Saline hydrides are compounds of hydrogen with most of the p-block elements.

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Solution:

Saline or ionic hydrides are compounds of hydrogen with most of the s-block metals. However, with p-block metals hydrogen forms molecular or covalent hydrides.

Q15: The bonding between hydrogen and other elements is always covalent. It just varies in its degrees of covalency. The general chemical formula of a hydride is MH_x . Here M is the other element with which hydrogen makes a binary compound. And the x denotes the number of hydrogen atoms. There are three possible types of hydrides based on their structure and chemical properties. Saline Hydrides form when hydrogen reacts with s-Block elements. These are elements found in Group 1 and Group 2. These alkali metals and alkaline earth metals are more electropositive than hydrogen. Covalent hydrides that form when hydrogen reacts with p-block elements. These are highly electronegative elements, far more than hydrogen. Hydrogen atoms form a covalent bond with these p-block atoms. Metallic Hydride are formed when hydrogen reacting with transition elements.

Assertion: Beryllium hydride is a covalent hydride.

Reason: The electronegativity difference between Be and H is very high.

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Solution:

Beryllium hydride is a covalent hydride, because the electronegativity difference between Be (1.5) and H (2.1) is small. Hence, Option "c" is the correct answer.

Q16: The bonding between hydrogen and other elements is always covalent. It just varies in its degrees of covalency. The general chemical formula of a hydride is MH_x . Here M is the other element with which hydrogen makes a binary compound. And the x denotes the number of hydrogen atoms. There are three possible types of hydrides based on their structure and chemical properties. Saline Hydrides form when hydrogen reacts with s-Block elements. These are elements found in Group 1 and Group 2. These alkali metals and alkaline earth metals are more electropositive than hydrogen. Covalent hydrides that form when hydrogen reacts with p-block elements. These are highly electronegative elements, far more than hydrogen. Hydrogen atoms form a covalent bond with these p-block atoms. Metallic Hydride are formed when hydrogen reacting with transition elements.

Assertion: The alkali metals can form ionic hydrides which contains the hydride ion H^- .

Reason: The alkali metals hydrides are good conductor of electricity.

- (A) Assertion and Reason both are correct statements and reason is correct explanation for assertion.
- (B) Assertion and Reason both are correct statements but reason is not correct explanation for assertion.
- (C) Assertion is correct statement but reason is wrong statement.
- (D) Assertion is wrong statement but reason is correct.

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Solution:

The alkali metal can form ionic hydrides which contains the hydride ion because the alkali metals have low electronegativity, their hydrides conduct electricity when fused.

Q17: Water has some very important uses in our world. It is the universal liquid and will find a use in 99% of life's functions. This unique ability of water to be so versatile and ubiquitous is due to some of its very impressive properties. These properties of water are often overlooked, but are the main reason for what makes water such a valuable compound. Chemical Formula of water as we are all aware is H_2O . The molecule of water has covalent bonding between Hydrogen and Oxygen atoms. Two hydrogen atoms form a bond with a single atom of oxygen. Water is colorless, odorless and tasteless liquid in its natural state. As we know, water has a boiling point of $100\text{ }^{\circ}\text{C}$. But this relatively high boiling point of water defies the trend in the periodic table.

Assertion: The viscosity of water is less than blood.

Reason: The viscosity of liquids decreases with increased in temperature.

- (A) Assertion and Reason both are correct statements and reason is correct explanation for assertion.
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Solution:

Whole blood has a much higher viscosity than water, as blood is thicker than water. Blood is more viscous than water. Also, the viscosity of the liquid decreases with a rise in temperature whereas it increases for the gases.

Q18: Water has some very important uses in our world. It is the universal liquid and will find a use in 99% of life's functions. This unique ability of water to be so versatile and ubiquitous is due to some of its very impressive properties. These properties of water are often overlooked, but are the main reason for what makes water such a valuable compound. Chemical Formula of water as we are all aware is H_2O . The molecule of water has covalent bonding between Hydrogen and Oxygen atoms. Two hydrogen atoms form a bond with a single atom of oxygen. Water is colorless, odorless and tasteless liquid in its natural state. As we know, water has a boiling point of $100\text{ }^{\circ}\text{C}$. But this relatively high boiling point of water defies the trend in the periodic table.

Assertion: Water has a low vapour pressure.

Reason: Water molecules form hydrogen bonds with other water molecules.

- (A) Assertion and Reason both are correct statements and reason is correct explanation for assertion.
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Assertion: Water has a low vapour pressure.

Reason: Water molecules form hydrogen bonds with other water molecules.

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Solution:

Water has exceptionally lower vapour pressure due to hydrogen bonding. Because of hydrogen-bonding less water vapour from as relatively few water molecules are capable of escaping the surface of the liquid. Hence, assertion and reason are correct and Reason is the correct explanation of Assertion. So, the correct answer is (a).

Q19: Water has some very important uses in our world. It is the universal liquid and will find a use in 99% of life's functions. This unique ability of water to be so versatile and ubiquitous is due to some of its very impressive properties. These properties of water are often overlooked, but are the main reason for what makes water such a valuable compound. Chemical Formula of water as we are all aware is H_2O . The molecule of water has covalent bonding between Hydrogen and Oxygen atoms. Two hydrogen atoms form a bond with a single atom of oxygen. Water is colorless, odorless and tasteless liquid in its natural state. As we know, water has a boiling point of $100\text{ }^{\circ}\text{C}$. But this relatively high boiling point of water defies the trend in the periodic table.

Assertion: The density of water remains constant as it is cooled from room temperature until its temperature reaches 4°C .

Reason: Below 4°C , the density decreases.

- (A) Assertion and Reason both are correct statements and reason is correct explanation for assertion.
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Assertion: The density of water remains constant as it is cooled from room temperature until its temperature reaches 4°C .

Reason: Below 4°C , the density decreases.

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Solution:

The volume of water decreases when water is cooled from room temperature, until its temperature reaches 4°C and hence density increases. Below 4°C , the volume increases and therefore the density decreases.

Q20: Water has some very important uses in our world. It is the universal liquid and will find a use in 99% of life's functions. This unique ability of water to be so versatile and ubiquitous is due to some of its very impressive properties. These properties of water are often overlooked, but are the main reason for what makes water such a valuable compound. Chemical Formula of water as we are all aware is H_2O . The molecule of water has covalent bonding between Hydrogen and Oxygen atoms. Two hydrogen atoms form a bond with a single atom of oxygen. Water is colorless, odorless and tasteless liquid in its natural state. As we know, water has a boiling point of $100\text{ }^{\circ}\text{C}$. But this relatively high boiling point of water defies the trend in the periodic table.

Assertion: Water in a beaker cannot be made to boil by placing it in a bath of boiling water.

Reason: Water in the beaker will be heated to $100\text{ }^{\circ}\text{C}$ but will not boil as for boiling it requires latent heat of steam which is not provided by water bath as both attains $100\text{ }^{\circ}\text{C}$.

- (A) Assertion and Reason both are correct statements and reason is correct explanation for assertion.
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Solution:

Water in a beaker cannot be made to boil by placing it in a bath of boiling water.

For water in a beaker to boil, it requires latent heat of vaporization which won't be provided by the bath of boiling water. Water in bath and water in a beaker will attain thermal equilibrium.

Hence, reason given is the correct explanation of the assertion. So, option (a) is correct.

Q21: Closely observe the structure of a water molecule. You will see one atom of oxygen and two atoms of hydrogen. Each atom of hydrogen bonds covalently with the atom of oxygen. So both atoms of hydrogen share one pair of electrons with the oxygen atom. Oxygen is a more electronegative element in comparison to hydrogen. This results in an uneven distribution of electron density. This gives the water molecule an angular bent structure. The H-O-H bond has an angle of about 104.5° . So it can be said that water molecule is polar. It has a slightly negative polarity near the oxygen atom and slightly positive charge near both the hydrogen atoms. The structure of the molecules of water in its frozen form i.e. ice is very unique. This very special hexagonal shape is what gives ice the unique property of being less dense than water. Since in the structure of ice there are empty spaces between the hexagonal structure, its density is less than that of water in its liquid state. This is why ice floats on water.

Assertion: H_2O is the only hydride of the chalcogens which is liquid.

Reason: In ice, each O atom is surrounded by 4H atoms.

- (A) Assertion and Reason both are correct statements and reason is correct explanation for assertion.
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Solution:

H₂O is only the hydride of chalcogens elements which is liquid whereas others are gases.

In ice, each H₂O molecule oxygen is held by two covalent bonds with hydrogen atoms and two hydrogen bonds with two other water molecules due to hydrogen bonding. Thus, one oxygen is bonded with 4 H-atoms. Hence, both assertion and reason are correct but reason is not the correct explanation of Assertion.

Q22: Closely observe the structure of a water molecule. You will see one atom of oxygen and two atoms of hydrogen. Each atom of hydrogen bonds covalently with the atom of oxygen. So both atoms of hydrogen share one pair of electrons with the oxygen atom. Oxygen is a more electronegative element in comparison to hydrogen. This results in an uneven distribution of electron density. This gives the water molecule an angular bent structure. The H-O-H bond has an angle of about 104.5° . So it can be said that water molecule is polar. It has a slightly negative polarity near the oxygen atom and slightly positive charge near both the hydrogen atoms. The structure of the molecules of water in its frozen form i.e. ice is very unique. This very special hexagonal shape is what gives ice the unique property of being less dense than water. Since in the structure of ice there are empty spaces between the hexagonal structure, its density is less than that of water in its liquid state. This is why ice floats on water.

Assertion: Ice cube floats on water.

Reason: Density of ice is less than that of water.

- (A) Assertion and Reason both are correct statements and reason is correct explanation for assertion.
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Solution:

When water freezes, water molecules form a crystalline structure maintained by hydrogen bonding. Solid water, or ice, is less dense than liquid water. Ice is less dense than water because the orientation of hydrogen bonds causes molecules to push farther apart, which lowers the density.

Q23: Closely observe the structure of a water molecule. You will see one atom of oxygen and two atoms of hydrogen. Each atom of hydrogen bonds covalently with the atom of oxygen. So both atoms of hydrogen share one pair of electrons with the oxygen atom. Oxygen is a more electronegative element in comparison to hydrogen. This results in an uneven distribution of electron density. This gives the water molecule an angular bent structure. The H-O-H bond has an angle of about 104.5° . So it can be said that water molecule is polar. It has a slightly negative polarity near the oxygen atom and slightly positive charge near both the hydrogen atoms. The structure of the molecules of water in its frozen form i.e. ice is very unique. This very special hexagonal shape is what gives ice the unique property of being less dense than water. Since in the structure of ice there are empty spaces between the hexagonal structure, its density is less than that of water in its liquid state. This is why ice floats on water.

Assertion: Water in liquid state is more stable than ice at room temperature.

Reason: Water in liquid form has higher entropy than ice.

- (A) Assertion and Reason both are correct statements and reason is correct explanation for assertion.
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Solution:

Fusion of ice is a spontaneous process, because the process again proceeds in the direction of more random state, i.e. liquid state is more random (higher entropy) in comparison to solid state, i.e., ice.

Q24: Closely observe the structure of a water molecule. You will see one atom of oxygen and two atoms of hydrogen. Each atom of hydrogen bonds covalently with the atom of oxygen. So both atoms of hydrogen share one pair of electrons with the oxygen atom. Oxygen is a more electronegative element in comparison to hydrogen. This results in an uneven distribution of electron density. This gives the water molecule an angular bent structure. The H-O-H bond has an angle of about 104.5° . So it can be said that water molecule is polar. It has a slightly negative polarity near the oxygen atom and slightly positive charge near both the hydrogen atoms. The structure of the molecules of water in its frozen form i.e. ice is very unique. This very special hexagonal shape is what gives ice the unique property of being less dense than water. Since in the structure of ice there are empty spaces between the hexagonal structure, its density is less than that of water in its liquid state. This is why ice floats on water.

Assertion: Bent shape is shown by water molecules.

Reason: Water molecules has bent molecular geometry.

- (A) Assertion and Reason both are correct statements and reason is correct explanation for assertion.
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Solution:

Bent shape is shown by water molecules. Lone pairs repel bonding electrons in water molecules. Both Assertion and Reason are correct and Reason is the correct explanation for Assertion. Water molecules has central O atom with 2 lone pairs and 2 bond pairs of electrons. O atom is sp^3 hybridized which results in tetrahedral electron pair geometry and bent molecular geometry.

Q25: While naturally existing, water dissolves many compounds in it. It changes some properties of water, making it 'hard' and 'soft.' The dissolution of calcium and magnesium salts in water leads to hydrogen carbonate, chloride, and sulfate, which ultimately makes the water 'Hard.' Hard water doesn't give lather with soap. Water-free from calcium and magnesium salt is called 'Soft' water, which gives lather easily with soaps. The hardness of the water could be completely removed easily by various techniques, either temporarily or permanently.

Assertion: Hard water is not suitable for laundry.

Reason: Soap containing sodium stearate reacts with hard water to precipitate out as calcium or magnesium salts.

- (A) Assertion and Reason both are correct statements and reason is correct explanation for assertion.
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Solution:

Hard water contains soluble calcium and magnesium salt like bicarbonates, chlorides and sulphates which reacts with the sodium stearate of soap and changes to the corresponding calcium and magnesium salt which precipitates out.

Q26: While naturally existing, water dissolves many compounds in it. It changes some properties of water, making it 'hard' and 'soft.' The dissolution of calcium and magnesium salts in water leads to hydrogen carbonate, chloride, and sulfate, which ultimately makes the water 'Hard.' Hard water doesn't give lather with soap. Water-free from calcium and magnesium salt is called 'Soft' water, which gives lather easily with soaps. The hardness of the water could be completely removed easily by various techniques, either temporarily or permanently.

Assertion: Soft water lathers with soap but not hard water.

Reason: Hard water reacts with soap, it form insoluble salts which form scum, not lather.

- (A) Assertion and Reason both are correct statements and reason is correct explanation for assertion.
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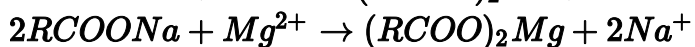
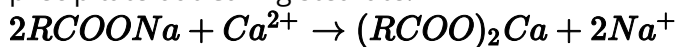
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- (D) Assertion is wrong statement but reason is correct.

Solution:

Hard water contains calcium and magnesium salts which react with soap containing sodium stearate ($C_{17}H_{35}COONa$) to precipitate out Ca/Mg stearate.



Q27: While naturally existing, water dissolves many compounds in it. It changes some properties of water, making it 'hard' and 'soft.' The dissolution of calcium and magnesium salts in water leads to hydrogen carbonate, chloride, and sulfate, which ultimately makes the water 'Hard.' Hard water doesn't give lather with soap. Water-free from calcium and magnesium salt is called 'Soft' water, which gives lather easily with soaps. The hardness of the water could be completely removed easily by various techniques, either temporarily or permanently.

Assertion: Temporary hardness can be removed by addition of lime.

Reason: $\text{Ca}(\text{HCO}_3)_2$ in hard water is converted to insoluble CaCO_3 on moderate heating.

- (A) Assertion and Reason both are correct statements and reason is correct explanation for assertion.
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Solution:

Temporary hardness can be reduced either by boiling the water, or by the addition of lime (calcium hydroxide) through the softening process of lime softening.

Q28: While naturally existing, water dissolves many compounds in it. It changes some properties of water, making it 'hard' and 'soft.' The dissolution of calcium and magnesium salts in water leads to hydrogen carbonate, chloride, and sulfate, which ultimately makes the water 'Hard.' Hard water doesn't give lather with soap. Water-free from calcium and magnesium salt is called 'Soft' water, which gives lather easily with soaps. The hardness of the water could be completely removed easily by various techniques, either temporarily or permanently.

Assertion: Boiling removes both temporary and permanent hardness from water.

Reason: Bicarbonates of calcium and magnesium undergoes decomposition on heating.

- (A) Assertion and Reason both are correct statements and reason is correct explanation for assertion.
- (B) Assertion and Reason both are correct statements but reason is not correct explanation for assertion.
- (C) Assertion is correct statement but reason is wrong statement.
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Q28: While naturally existing, water dissolves many compounds in it. It changes some properties of water, making it 'hard' and 'soft.' The dissolution of calcium and magnesium salts in water leads to hydrogen carbonate, chloride, and sulfate, which ultimately makes the water 'Hard.' Hard water doesn't give lather with soap. Water-free from calcium and magnesium salt is called 'Soft' water, which gives lather easily with soaps. The hardness of the water could be completely removed easily by various techniques, either temporarily or permanently.

Assertion: Boiling removes both temporary and permanent hardness from water.

Reason: Bicarbonates of calcium and magnesium undergoes decomposition on heating.

- (A) Assertion and Reason both are correct statements and reason is correct explanation for assertion.
- (B) Assertion and Reason both are correct statements but reason is not correct explanation for assertion.
- (C) Assertion is correct statement but reason is wrong statement.
- (D) Assertion is wrong statement but reason is correct.

Solution:

Boiling removes only temporary hardness because soluble bicarbonates of calcium and magnesium undergo decomposition on heating and insoluble carbonates are formed.

Q29: Hydrogen is the first element in the periodic table and has the atomic number one. Those elements which have the same atomic number but a different mass number are called isotopes. There are three isotopes of hydrogen namely, protium ${}_1\text{H}^1$, deuterium ${}_1\text{H}^2$ or D and lastly tritium ${}_1\text{H}^3$ or T. The isotopes are different because of the different number of neutrons present in them. In protium, there is no presence of neutrons, whereas in deuterium we have one neutron and in tritium, we have two neutrons. The most prominent form of hydrogen is protium, 0.0156% of hydrogen is present on the earth's surface as deuterium. In tritium, the concentration is one atom per 10^{18} atoms of protium. As the electronic configuration of isotopes is the same, they all have similar chemical properties. But they have a difference in their rates of reaction, this happens because of the different bond dissociation enthalpies. They have different physical properties because of the large variations in mass.

Assertion: All the three isotopes of hydrogen have almost the same chemical properties.

Reason: Isotopes differ from one another in respect of the presence of neutrons.

- (A) Assertion and Reason both are correct statements and reason is correct explanation for assertion.
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Solution:

Isotope have almost the same chemical properties because they have the same electronic configuration. All three isotopes of hydrogen have 1 electron in their valence shell. ${}^1_1\text{H}$, ${}^2_1\text{H}$ and ${}^3_1\text{H}$ have 0, 1 and 2 neutrons respectively.

Q30: Hydrogen is the first element in the periodic table and has the atomic number one. Those elements which have the same atomic number but a different mass number are called isotopes. There are three isotopes of hydrogen namely, protium ${}_1\text{H}^1$, deuterium ${}_1\text{H}^2$ or D and lastly tritium ${}_1\text{H}^3$ or T. The isotopes are different because of the different number of neutrons present in them. In protium, there is no presence of neutrons, whereas in deuterium we have one neutron and in tritium, we have two neutrons. The most prominent form of hydrogen is protium, 0.0156% of hydrogen is present on the earth's surface as deuterium. In tritium, the concentration is one atom per 1018 atoms of protium. As the electronic configuration of isotopes is the same, they all have similar chemical properties. But they have a difference in their rates of reaction, this happens because of the different bond disassociation enthalpies. They have different physical properties because of the large variations in mass.

Assertion: ${}_1\text{H}^1$, ${}_1\text{H}^2$ and ${}_1\text{H}^3$ are isotopes of hydrogen.

Reason: Nuclides of the same element of different mass numbers are called isotopes of that element.

- (A) Assertion and Reason both are correct statements and reason is correct explanation for assertion.
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Solution:

Hydrogen (${}^1\text{H}$) has three naturally occurring isotopes, sometimes denoted ${}^1\text{H}$, ${}^2\text{H}$ and ${}^3\text{H}$. The ${}^2\text{H}$ (or hydrogen-2) isotope is deuterium and the ${}^3\text{H}$ (or hydrogen-3) isotope is tritium. The symbols D and T are sometimes used for deuterium and tritium respectively. Here, 1, 2, and 3 represent the mass number of hydrogen, deuterium, and tritium respectively.

All isotopes of a given element have the same number of protons but different numbers of neutrons in each atom. The number of nuclides (both protons and neutrons) in the nucleus is the atom's mass number, and each isotope of a given element has a different mass number.

Hence, both assertion and reason are correct and the reason is the correct explanation for the assertion.

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Assertion: The relative proportion of Tritium and Protium is 1: 10^{17} .

Reason: The property responsible for the low proportion of Tritium is its non-radioactivity.

- (A) Assertion and Reason both are correct statements and reason is correct explanation for assertion.
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Solution:

The relative proportion of Tritium and Protium is $1: 10^{17}$. Tritium is a radioactive element, this property responsible for the low proportion of tritium.

Q32: Hydrogen is the first element in the periodic table and has the atomic number one. Those elements which have the same atomic number but a different mass number are called isotopes. There are three isotopes of hydrogen namely, protium ${}_1\text{H}^1$, deuterium ${}_1\text{H}^2$ or D and lastly tritium ${}_1\text{H}^3$ or T. The isotopes are different because of the different number of neutrons present in them. In protium, there is no presence of neutrons, whereas in deuterium we have one neutron and in tritium, we have two neutrons. The most prominent form of hydrogen is protium, 0.0156% of hydrogen is present on the earth's surface as deuterium. In tritium, the concentration is one atom per 10^{18} atoms of protium. As the electronic configuration of isotopes is the same, they all have similar chemical properties. But they have a difference in their rates of reaction, this happens because of the different bond dissociation enthalpies. They have different physical properties because of the large variations in mass.

Assertion: Tritium is used as tracer element in preference to deuterium.

Reason: Tritium is non-toxic and it emits low energy beta radiation.

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Solution:

Tritium is used as a radioactive tracer element in biological research in the form of tritiated water (H_2O or T_2O) and also used in capsules surrounded by a fluorescing compound (e.g., zinc sulfide) to provide illumination which must be independent of the electricity supply. For example, it is used in emergency exit signs, self-luminous airport runway and helicopter pad lights, and light wands for use in directing traffic.

Q33: Heavy water is deuterium oxide i.e., D_2O . Heavy water contains a deuterium atom instead of that of hydrogen. In fact, deuterium is often referred to as Heavy Hydrogen and while it looks exactly like normal water, heavy water has a few different chemical and physical properties.

The name heavy water comes from the fact that it is quite simply heavier than water. Deuterium is almost twice as heavy as protium and also less stable. D_2O has the same qualities as water of being tasteless and odorless. However, its boiling point is slightly heavier than that of ordinary water.

Assertion: D_2O is called heavy water.

Reason: Its degree of dissociation is high.

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Solution:

D_2O is called heavy water. Due to stronger D-O bonds, the degree of dissociation of D_2O is lower than that of H_2O .

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Assertion: Heavy water preferred over ordinary water as a moderator in reactors.

Reason: Heavy water, used for slowing down the neutrons, has lesser absorption probability of neutrons than ordinary water.

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Solution:

A bombarding particle gives up most energy when it makes an elastic collision with a particle of similar mass. For neutrons, hydrogen atoms would be most effective but unfortunately absorption occurs. However, deuterium (heavy water) and carbon (as graphite) are both suitable in which less absorption occurs. Therefore, they are used as moderators.

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Assertion: NaCl is less soluble in heavy water than ordinary water.

Reason: Dielectric constant of ordinary water is more than that of heavy water.

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Solution:

The dielectric constant of ordinary water (81) is greater than that of heavy water (80), hence NaCl is less soluble in heavy water.

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Assertion: The reactions of heavy water are slow.

Reason: Heavy water is of higher mass.

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Solution:

As Deuterium contains one extra neutron, so heavy water has higher molecular mass than water and because of that it is less reactive.

A molecule of heavy water has two deuterium atoms in place of the two protium atoms of ordinary "light" water. The weight of a heavy water molecule, however, is not substantially different from that of a normal water molecule.

Q37: In what respect electronic configuration of hydrogen and halogens are similar?

- (A) Hydrogen and halogens have one electron in their outermost shell.
- (B) Hydrogen and halogens have one electron less than the noble gas configuration.
- (C) Hydrogen and halogens can lose one electron to form positive ions.
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Solution:

H – $1s^1$; He – $1s^2$

X – ns^2np^5 ; Ne – $1s^22s^22p^6$

Both have one electron less than the nearest noble gas configuration.

Q38: Which of the following properties of hydrogen is incorrect?

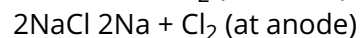
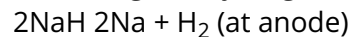
- (A) Like halogens, hydrogen exists as a diatomic gas.
- (B) Like halogens, hydrogen exhibits -1 oxidation state in its compounds with metals.
- (C) Like halogens, hydrogen is liberated at cathode.
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Solution:

Like halogens, hydrogen is liberated at anode e.g.,



Q39: In which of the following properties hydrogen does not show similarity with alkali metals?

- (A) Electropositive character
- (B) Reducing nature
- (C) Electronic configuration (ns^1)
- (D) Diatomic nature of molecule

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Solution:

Hydrogen forms a diatomic molecule similar to halogens but alkali metals are monoatomic

Q40: A deuterium is

- (A) an electron with a positive charge
- (B) a nucleus having two protons
- (C) a nucleus containing a neutron and two protons
- (D) a nucleus containing a neutron and a proton.

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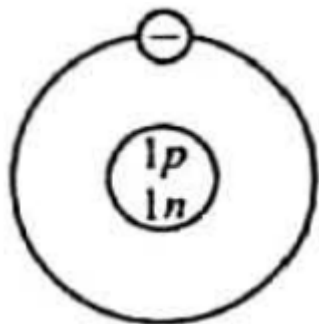
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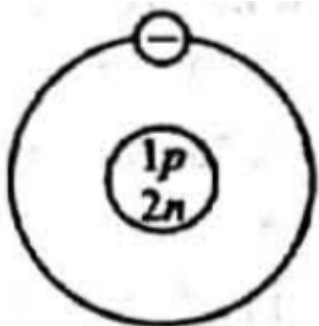
Deuterium atomic number is 1 and mass number is 2. It has 1 electron, 1 proton, 1 neutron.

Q41: Which of the following is an atom of tritium?

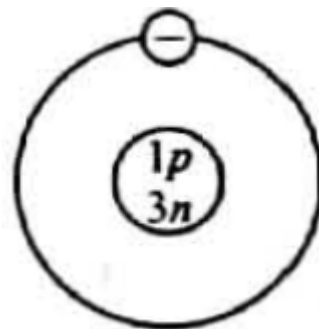
(A)



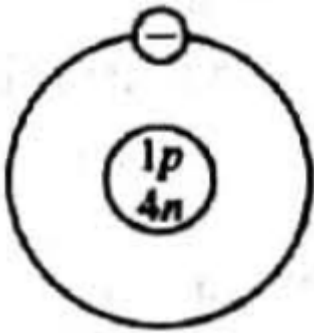
(B)



(C)

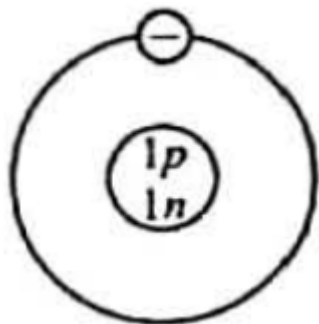


(D)

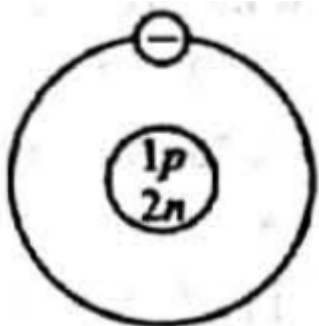


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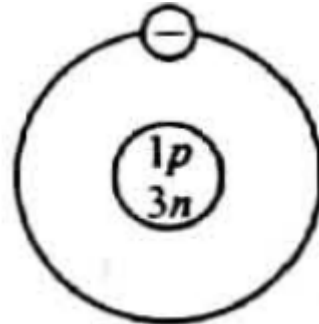
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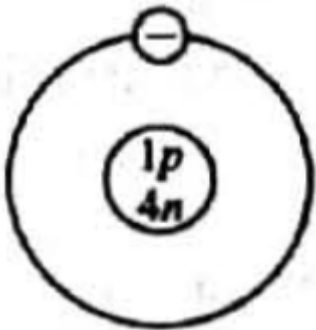
(B)



(C)



(D)



Solution:

Tritium is ${}^3_1\text{H}$ having one proton and two neutrons. It has number of protons = 1, number of electrons = 1, number of neutrons = 2.

Q42: The isotopes of hydrogen have different physical properties due to difference in mass. They have almost same chemical properties with a difference in their rates of reactions which is mainly due to

- (A) Their different enthalpy of bond dissociation
- (B) Different electronic configurations
- (C) Different atomic masses
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Solution:

Enthalpy of bond dissociation of deuterium is higher ($443.35 \text{ kJ mol}^{-1}$) than hydrogen ($435.88 \text{ kJ mol}^{-1}$).

Q43: What is the trend of boiling point of hydrides of **N**, **O** and **F**?

- (A) Due to lower molecular masses NH_3 , H_2O and HF have lower boiling points than those that of the subsequent group members hydrides.
- (B) Due to higher electronegativity of N, O and F, NH_3 , H_2O and HF show hydrogen bonding and hence higher boiling points than the hydrides of their subsequent group members.
- (C) There is no regular trend in boiling points of hydrides.
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Solution:

Due to higher electronegativity, the extent of hydrogen bonding is quite appreciable.

Q44: On moving left to right in period what is the order of acidic character of hydrides?

- (A) $\text{NH}_3 < \text{H}_2\text{O} < \text{HF}$
- (B) $\text{HF} < \text{H}_2\text{O} < \text{NH}_3$
- (C) $\text{H}_2\text{O} < \text{HF} < \text{NH}_3$
- (D) $\text{H}_2\text{O} < \text{NH}_3 < \text{HF}$

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- (D) $\text{H}_2\text{O} < \text{NH}_3 < \text{HF}$

Solution:

On moving from left to right in a period, acidic character of hydrides increase. NH_3 is basic, H_2O is neutral and HF is acidic.

Q45: In complex hydrides, hydride ions act as ligand and are coordinated to metal ions. These hydrides are good reducing agents. Which of the following hydrides is not a complex hydride?

- (A) LiAlH_4
- (B) NaBH_4
- (C) $(\text{AlH}_3)_n$
- (D) LiBH_4

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- (C) $(\text{AlH}_3)_n$
- (D) LiBH_4

Solution:

$(\text{AlH}_3)_n$ is a polymeric hydride like $(\text{BeH}_2)_n$, $(\text{MgH}_2)_n$, etc.

Q46: Which of the following statements regarding hydrides is not correct?

- (A) Ionic hydrides are crystalline, non-volatile and non-conducting in solid state.
- (B) Electron-deficient hydrides act as Lewis acids or electron acceptors.
- (C) Elements of group -13 form electron deficient hydrides.
- (D) Elements of group 15-17 form electron precise hydrides.

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Solution:

Elements of group 15-17 form electron-rich hydrides. Group 14 elements form electron-precise hydrides.

Q47: Carbon hydrides of type, C_nH_{2n+1} do not act as Lewis acid or Lewis base. They behave as normal covalent hydrides because

- (A) carbon hydrides are electron rich hydrides.
- (B) carbon hydrides are electron-deficient hydrides.
- (C) carbon hydrides are electron-precise hydrides.
- (D) carbon hydrides are non-stoichiometric hydrides.

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- (D) carbon hydrides are non-stoichiometric hydrides.

Solution:

Carbon hydrides with general formula C_nH_{2n+1} are electron-precise hydrides like CH_4 . They have a complete octet hence they do not behave as Lewis acid or Lewis base.

Q48: From group 6 only one metals forms hydride. The metal is

- (A) Mo
- (B) W
- (C) Cr
- (D) Co

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- (A) Mo
- (B) W
- (C) Cr
- (D) Co

Solution:

Only chromium from group 6 forms CrH .

Q49: Phosphorus cannot form PH_5 with its outer electronic configuration as $3s^2 3p^3$ because

- (A) phosphorus cannot show +5 oxidation state
- (B) PH_5 is not a stable compound
- (C) $\Delta_{\alpha} \text{H}$ value of dihydrogen and $\Delta_{eg} \text{H}$ value of hydrogen do not favour higher oxidation state of phosphorus
- (D) phosphorus is not very reactive hence does not form PH_5 .

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- (D) phosphorus is not very reactive hence does not form PH_5 .

Solution:

Since higher oxidation state of PH_5 not favoured by high $\Delta_{\alpha} \text{H}$ of dihydrogen and $\Delta_{eg} \text{H}$ of hydrogen, P does not form PH_5 though it exhibits +3 and +5 oxidation states.

Q50: Among NH_3 , H_2O , HF and H_2S which would have highest magnitude of hydrogen bonding?

- (A) HF due to maximum polarity.
- (B) H_2O due to lone pairs of electrons.
- (C) NH_3 due to small size of nitrogen.
- (D) H_2S due to higher electron affinity of sulphur.

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Solution:

Since H - F bond has maximum polarity due to most electronegative nature of fluorine, hydrogen bonding is maximum in HF molecules.

Q51: Non-stoichiometric hydrides are produced by

- (A) palladium, vanadium
- (B) manganese, lithium
- (C) nitrogen, fluorine
- (D) carbon, nickel

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- (A) palladium, vanadium
- (B) manganese, lithium
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Solution:

Non-stoichiometric hydrides are formed by d-block and f-block element.

Q52: Given below are the elements and the type of hydrides formed by them. Mark the incorrect match.

- (A) Phosphorus-Molecular hydride
- (B) Potassium-Ionic hydride
- (C) Vanadium-Interstitial hydride
- (D) Nitrogen-Electron-deficient covalent hydride

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- (C) Vanadium-Interstitial hydride
- (D) Nitrogen-Electron-deficient covalent hydride

Solution:

Nitrogen forms electron-rich covalent molecular hydrides.

Q53: Liquid water is denser than ice due to

- (A) higher surface tension
- (B) hydrogen bonding
- (C) van der Waals forces
- (D) covalent bonding

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Solution:

Due to extensive H-bonding water is denser than ice. Some of the H-bonding breaks when ice is formed.

Q54: The density of water is less in its solid state because

- (A) in solid state (ice), water molecules are arranged in highly order open cage like structure
- (B) more extensive hydrogen bonding is present in solid state ice
- (C) the water molecules are closest in solid state of water
- (D) water in rigid crystalline, closely packed structure in its solid state

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- (B) more extensive hydrogen bonding is present in solid state ice
- (C) the water molecules are closest in solid state of water
- (D) water in rigid crystalline, closely packed structure in its solid state

Solution:

In solid state (ice), water molecules are arranged in highly order open cage like structure.

ρ of ice $<$ ρ of water

Q55: Water plays a key role in the biosphere. It is due to certain properties of H_2O as compared to other liquids. These are except

- (A) higher specific heat
- (B) lesser thermal conductivity
- (C) high dielectric constant
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- (A) higher specific heat
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Solution:

Water has higher thermal conductivity, specific heat, dielectric constant and surface tension in comparison to other liquids.

Q56: The H-O-H angle in water molecule is about

- (A) **90°**
- (B) **180°**
- (C) **102°**
- (D) **105°**

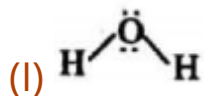
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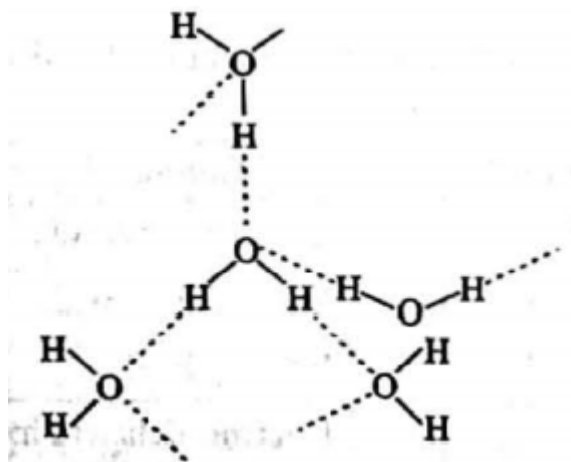
Solution:

H - O - H angle in water is slightly less than the typical tetrahedral angle. It is 104.5° .

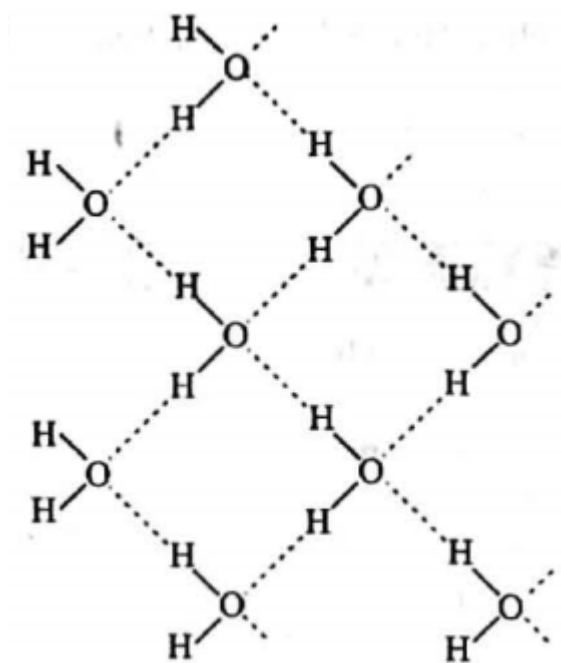
Q57: Choose the correct statement about the given figures.



(II)

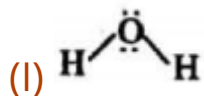


(III)

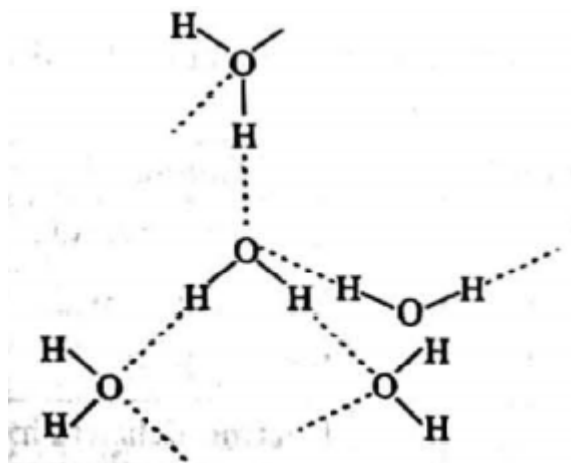


- (A) (II) represents solid state while (III) represents liquid state
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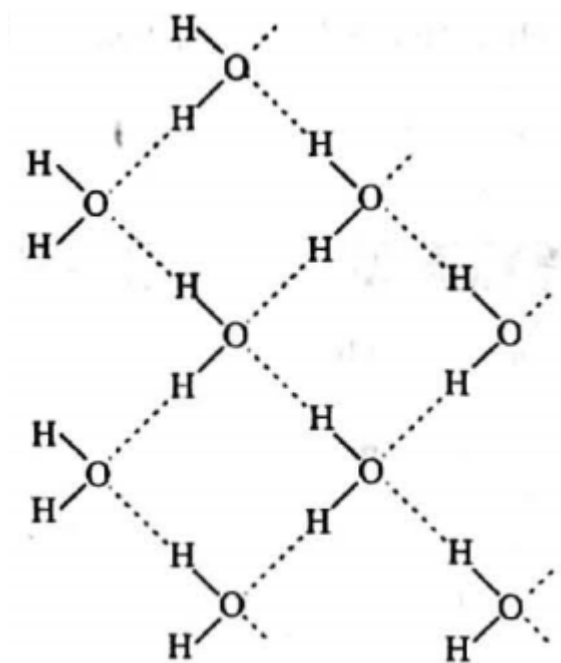
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- (C) (I) represents solid state while (III) represents liquid state
- (D) (I) represents liquid state while (III) represents solid state.

Solution:

In liquid state water molecules are hydrogen bonded while in solid state water molecules are arranged in tetrahedral manner with open cage structure.

Q58: Which is not a property of water?

- (A) It is a colourless and tasteless liquid.
- (B) There is no hydrogen bonding in solid state of water.
- (C) It is an excellent solvent for transportation of ions in plants and animals.
- (D) Frozen water is lighter than liquid water.

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- (C) It is an excellent solvent for transportation of ions in plants and animals.
- (D) Frozen water is lighter than liquid water.

Solution:

There is extensive hydrogen bonding in solid state of water.

Q59: Which of the statements given below are true for the structure of water molecule?

(i) Oxygen undergoes sp^3 hybridisation.

(ii) Due to presence of two lone pairs of electrons on oxygen the H - O - H bond angle is 118.4° .

(iii) Due to angular geometry the net dipole moment of water is not zero, $\mu = 1.84 \text{ D}$.

(A) (i) and (ii)

(B) (ii) and (iii)

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(A) (i) and (ii)

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Solution:

Due to presence of two lone pairs of electrons on oxygen atom the bond angle H - O - H is 104.5° .

Q60: The maximum number of hydrogen bonds formed by a water molecule in ice is

- (A) 4
- (B) 1
- (C) 2
- (D) 3

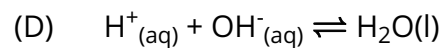
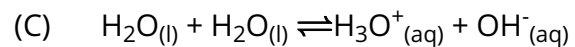
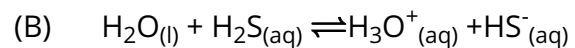
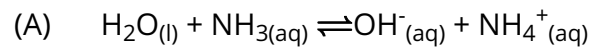
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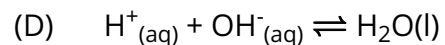
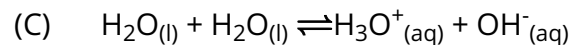
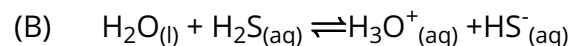
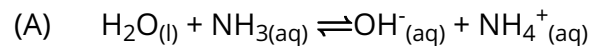
Solution:

Each water molecule can form a maximum of 4 hydrogen bonds with neighbouring molecule

Q61: In which of the following reactions H_2O acts as a Bronsted acid?

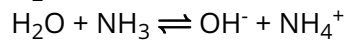


Q61: In which of the following reactions H_2O acts as a Bronsted acid?



Solution:

H_2O acts as a Bronsted acid and gives a proton to react with a base.



In $\text{H}_2\text{O} + \text{H}_2\text{S} \rightarrow \text{H}_3\text{O}^{+} + \text{HS}^{-}$, H_2O act as a base with H_2S .

Q62: What is the reaction given below called?



- (A) Hydrolysis of water
- (B) Hydration of water
- (C) Diproportionation of water
- (D) Auto-protolysis of water

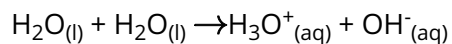
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- (A) Hydrolysis of water
- (B) Hydration of water
- (C) Diproportionation of water
- (D) Auto-protolysis of water

Solution:

The reaction in which two molecules of the same species react to give ions by proton transfer is known as auto-protolysis or self ionisation. In this reaction, proton hydrogen from one H_2O molecule is transferred to another molecule.



Q63: Given below are two reactions shows reduction of water with sodium and carbon dioxide. What is the nature of water in these reactions?



- (A) In (ii) water acts as an oxidising agent and in (i) it acts as a reducing agent.
- (B) In (i) water acts as an oxidising agent while in (ii) it acts as a reducing agent.
- (C) In both, (i) and (ii) hydrogen acts as a reducing agent.
- (D) In both, (i) and (ii) hydrogen acts as an oxidising agent.

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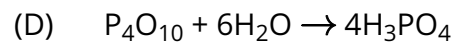
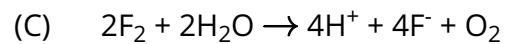
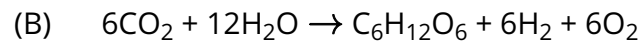
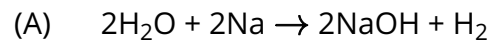


- (A) In (ii) water acts as an oxidising agent and in (i) it acts as a reducing agent.
- (B) In (i) water acts as an oxidising agent while in (ii) it acts as a reducing agent.
- (C) In both, (i) and (ii) hydrogen acts as a reducing agent.
- (D) In both, (i) and (ii) hydrogen acts as an oxidising agent.

Solution:

With sodium metal water acts as an oxidizing agent and gets reduced to H_2 , with CO_2 water acts as a reducing agent and is oxidized to oxygen.

Q64: Which of the following reactions shows reduction of water?



Q64: Which of the following reactions shows reduction of water?

- (A) $2\text{H}_2\text{O} + 2\text{Na} \rightarrow 2\text{NaOH} + \text{H}_2$
- (B) $6\text{CO}_2 + 12\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{H}_2 + 6\text{O}_2$
- (C) $2\text{F}_2 + 2\text{H}_2\text{O} \rightarrow 4\text{H}^+ + 4\text{F}^- + \text{O}_2$
- (D) $\text{P}_4\text{O}_{10} + 6\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{PO}_4$

Solution:

Water is reduced to dihydrogen by highly electropositive metals. In this reaction H_2O is reduced to H_2 . In other two reactions, H_2O is oxidized to O_2 and P_4O_{10} is hydrolysed by water.

Q65: Study the following reactions and mark the correct properties shown by water



- (A) All oxides react with water to give hydroxides.
- (B) Acidic oxides are formed by metals and basic oxides by non-metals.
- (C) Non-metal oxides combine with water to form acids while metallic oxides combine with water to form alkalies.
- (D) Acidic oxides are stronger than basic oxides since they form strong acids.

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- (D) Acidic oxides are stronger than basic oxides since they form strong acids.

Solution:

Acidic oxides form acids while basic oxides form alkalies with water.

Q66: How many hydrogen bonded water molecules are associated with $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$?

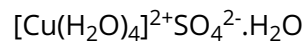
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- (C) Four
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- (A) Five
- (B) One
- (C) Four
- (D) Three

Solution:

Only one water molecule which is outside the coordination sphere is hydrogen bonded. Other four molecules of H_2O are coordinated.



Q67: During hydrate formation from aqueous solution, water can be associated in different forms. Indicate the wrong combination.

(i) Coordinated water – $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}3\text{Cl}^-$

(ii) Interstitial water – $\text{BaCl}_2 \cdot 2\text{H}_2\text{O}$

(iii) Hydrogen bonded water – $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}\text{SO}_4^{2-} \cdot \text{H}_2\text{O}$

(A) (i)

(B) (ii)

(C) (iii)

(D) None of these

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(A) (i)

(B) (ii)

(C) (iii)

(D) None of these

Solution:

All combinations are correct.

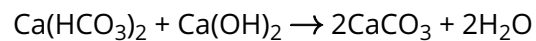
Q68: The temporary hardness of water due to calcium bicarbonate can be removed by adding

- (A) CaCO_3
- (B) CaCl_2
- (C) HCl
- (D) Ca(OH)_2

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Solution:



Q69: In a permutit, the calcium and magnesium ions of hard water are exchanged by

- (A) CO_3^{2-} and HCO_3^- ions of permutit
- (B) Na^+ ions of permutit
- (C) Al^{3+} ions of permutit
- (D) Si^{4+} ions of permutit

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Solution:



Q70: Which of the following represent calgon?

- (A) $\text{Na}_2\text{Al}_2\text{Si}_2\text{O}_8$
- (B) $\text{Mg}_3(\text{PO}_4)_2$
- (C) $\text{Na}_2[\text{Na}_4(\text{PO}_2)_6]$
- (D) $\text{Na}_2[\text{Mg}_2(\text{PO}_3)_4]$

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- (C) $\text{Na}_2[\text{Na}_4(\text{PO}_2)_6]$
- (D) $\text{Na}_2[\text{Mg}_2(\text{PO}_3)_4]$

Solution:

The complex salt of metaphosphoric acid, sodium hexametaphosphate (NaPO_3)₆ is known as calgon. It is represented as $\text{Na}_2[\text{Na}_4(\text{PO}_3)_6]$.

Q71: The process used for the removal of hardness of water is

- (A) Baeyer
- (B) Calgon
- (C) Hoope
- (D) Serpeck

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Solution:

Calgon is process used for the removal of hardness of water

Q72: A water sample is said to contain permanent hardness if water contains

- (A) sulphates and chlorides of calcium and magnesium
- (B) carbonates of calcium and magnesium
- (C) bicarbonates of calcium and magnesium
- (D) sulphates and chlorides of sodium and potassium.

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- (C) bicarbonates of calcium and magnesium
- (D) sulphates and chlorides of sodium and potassium.

Solution:

Permanent hardness of water is due to sulphates and chlorides of calcium and magnesium.

Q73: The formula for permutit or zeolite which is used as softner in ion-exchange method is

- (A) NaSiO_4
- (B) NaAlO_2
- (C) $\text{Ca}_3(\text{PO}_4)_2$
- (D) Na_2SO_4

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- (C) $\text{Ca}_3(\text{PO}_4)_2$
- (D) Na_2SO_4

Solution:

Zeolite/permutit is hydrated sodium aluminium silicate which is also written as NaZ

Q74: Which of the following is not a disadvantage of using hard water?

- (A) In production of steam in boilers.
- (B) Formation of scales in cooking utensils
- (C) In cooking, bathing and washing
- (D) In ion exchangers

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- (B) Formation of scales in cooking utensils
- (C) In cooking, bathing and washing
- (D) In ion exchangers

Solution:

In an ion exchanger using hard water is not a disadvantage since it becomes soft by exchanging anions or cations.

Q75: Polyphosphates like sodium hexametaphosphate (calgon) are used as water softening agents because they

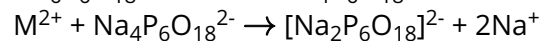
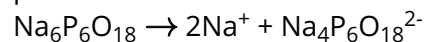
- (A) form soluble complexes with anionic species.
- (B) precipitate anionic species
- (C) form soluble complexes with cationic species.
- (D) precipitate cationic species.

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- (A) form soluble complexes with anionic species.
- (B) precipitate anionic species
- (C) form soluble complexes with cationic species.
- (D) precipitate cationic species.

Solution:

Polyphosphates like sodium hexametaphosphate(calgon) form soluble complexes with cations like Ca^{2+} and Mg^{2+} present in hard water and hence used as water softening agents.



Q76: What is meant by demineralised water?

- (A) Water free from cations and anions.
- (B) Water free from minerals dissolved in it.
- (C) Water free from impurities.
- (D) Water free from Na^+ and K^+ ion

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Solution:

Water free from cations (Ca^{2+} , M^{2+} , etc.) and anions (Cl^- , SO_4^{2-} , HCO_3^- , etc.) is called deionized or demineralised water

Q77: Heavy water is obtained by

- (A) boiling water
- (B) heating H_2O_2
- (C) prolonged electrolysis of H_2O
- (D) All of these

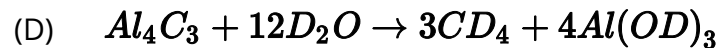
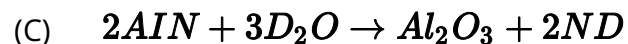
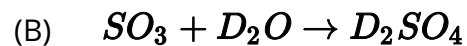
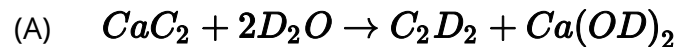
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Solution:

D_2O is prepared by repeated electrolysis of ordinary water which contains a small amount of alkali.

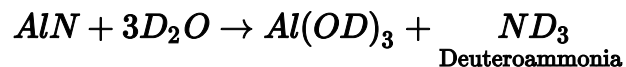
Q78: Which of the following reactions is not used in preparation of deuterium compounds using heavy water?



Q78: Which of the following reactions is not used in preparation of deuterium compounds using heavy water?

- (A) $CaC_2 + 2D_2O \rightarrow C_2D_2 + Ca(OD)_2$
- (B) $SO_3 + D_2O \rightarrow D_2SO_4$
- (C) $2AlN + 3D_2O \rightarrow Al_2O_3 + 2ND_3$
- (D) $Al_4C_3 + 12D_2O \rightarrow 3CD_4 + 4Al(OD)_3$

Solution:



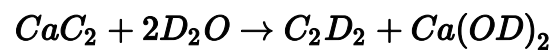
Q79: Which compound is formed when calcium carbide reacts with heavy water?

- (A) C_2D_2
- (B) CaD_2
- (C) CD_2
- (D) Ca_2D_2

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- (A) C_2D_2
- (B) CaD_2
- (C) CD_2
- (D) Ca_2D_2

Solution:



Q80: Some of the major uses of heavy water are given below. Which one is not correct?

- (A) It is used as a moderator in nuclear reactors
- (B) It is used as a tracer compound for studying reaction mechanism
- (C) High concentration of heavy water accelerates the growth of plants
- (D) It is used in preparing deuterium

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- (A) It is used as a moderator in nuclear reactors
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- (C) High concentration of heavy water accelerates the growth of plants
- (D) It is used in preparing deuterium

Solution:

High concentration of heavy water retards the growth of the plants.

